



Name: _____

Class: _____

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Programming 4 — Operator Activities

Work down the page. You are not expected to reach the bottom.

Warm up

1 • Predict, then run adapted from Paul Long

Write down your answer *before* you touch a keyboard.

Expression	Your prediction	What it gave
80 DIV 11		
80 MOD 11		
11 DIV 80		
11 MOD 80		

Rows 3 and 4 are the interesting ones. Swapping the numbers round does not swap the answers round.

Before you go further — loops and decisions

Most of what follows needs a loop. Here is the review, because it is easy to mix these two up.

The difference is how many times the question gets asked.

if — asks ONCE

```
if age >= 17:
    print("Old enough")
else:
    print("Not old enough")
```

The question is asked, one branch runs, and the program moves on. **It never comes back.**

while — asks OVER AND OVER

```
while age < 17:
    print("Too young")
    age = age + 1
```

The question is asked, the block runs, and then **it goes back and asks again**. It only stops when the answer is finally no.

The rest of the family, briefly:

Keyword	What it does
if	Asks a question once. Runs its block if the answer is yes.
elif	“Otherwise, try <i>this</i> question.” Only reached if the ones above it were no.
else	“Otherwise do this.” No question — it is the catch-all, and it is optional.
while	Asks a question, runs its block, then asks the same question again .
for	Repeats a known number of times. Use it when you can count in advance.

Every while loop needs three things. Miss one and it breaks.

1. Something set up BEFORE the loop — the variable the question is about has to already exist, or the very first question cannot be asked.

2. A question at the top — something that is either true or false.

3. Something INSIDE that changes the answer. This is the one people forget. If nothing in the block changes the thing being tested, the answer stays yes for ever and the program never stops. That is an **infinite loop**, and the only way out is to close the tab.

Before you run any while loop, point at the line inside it that will eventually make the question false. If you cannot find that line, do not run it.

for or while? Ask yourself: *do I know how many times before I start?* If yes, for. If you only know when to *stop*, while.

Build

2 • Pulling digits out of a number take this one slowly

MOD 10 gives you the last digit of a number. DIV 10 throws that last digit away. Between them, you can take any number completely apart.

```
number = 4827
print(number % 10)      # 7    -- the last digit
print(number // 10)     # 482  -- everything except the last digit
```

(a) Print the last digit of 4827. _____

(b) Print the *second*-from-last digit. _____

Hint: strip one digit off first, then take the last digit of what is left.

(c) Print the SUM of all four digits of 4827. [the long one]

This one is worth doing properly, because the pattern in it turns up everywhere. Work through the steps in order and do not skip to the code.

Step 1 — do it by hand first. Fill in this table. The trick is that whatever // 10 leaves behind becomes the number you are holding on the next row.

Number I am holding	% 10 gives me	// 10 leaves
4827	7	482
482		

The middle column is your four digits. Add them up — that is the answer you are aiming for.

Step 2 — three questions about your table.

- (i) In each row you did the *same two things*. What were they?
- (ii) **How did you know to stop?** You could see 4827 has four digits — but the program cannot. *What was true about the number at the bottom that was not true before?*
- (iii) You were adding the digits up as you went. Where did that total live? It was not in the original number — you had to *start* it somewhere and update it each row.

Step 3 — try it with 57 before you write anything.

57 takes **two** passes to reach 0. 4827 takes four. 123456 takes six. **So you cannot write “repeat four times”** — the program is never told how many digits there are. It only knows the number it is holding right now.

That is exactly why this is a while loop and not a for loop.

Step 4 — fill in the four blanks.

```

number = 4827
total = 0

while _____ :           # is there anything left to strip?
    digit = _____       # the middle column of your table
    total = _____       # add that digit to the running total
    number = _____      # the right-hand column -- what is left

print(total)

```

Two of these you have already written — digit is part (a) and number is part (b), just with the word number in place of 4827, because it changes every pass.

The line that looks wrong: `total = total + digit`

As maths that is nonsense — nothing equals itself plus something. **But = does not mean “equals”.** It means *put this into that*. Work out the right-hand side first, then drop the answer into the box on the left.

First pass: total is 0 and digit is 7, so the right-hand side is 7, and 7 goes into total. Second pass: total is 7 and digit is 2, so 9 goes into total.

This is called a **running total**. Counting, totalling, averaging — all the same three lines. You will use it constantly.

The bug to avoid — and it is the self-checkout bug from Programming 2.

`total = 0` must sit **above** the loop, not inside it. Indent it into the loop and it resets to zero on every single pass, so you finish with only the last digit.

You would get 4 instead of 21 — and there is **no error message**. The program runs perfectly and gives you the wrong answer. That is a logic error.

Try it both ways on purpose. Seeing it fail once is worth more than being told.

Step 5 — when it works, test it. Try 57 (should give 12), 9 (should give 9) and 1000 (should give 1). **Then try 0.** What happens, and is that right?

3 • A proper stopwatch

3661 seconds is 1 hour, 1 minute and 1 second. Write a program that turns any number of seconds into hours, minutes and seconds.

Ask the user for a number of seconds. Print it like 1h 1m 1s.

Test it with: 60 · 3599 · 3600 · 3661 · 86399.

4 • The change machine

A vending machine works in pence. Ask the user for an amount in pence and print it properly.

287 should print as 2 pounds and 87 pence.

Then make it harder: give the change in coins. How many £1, 50p, 20p, 10p, 5p, 2p and 1p? *Each coin is one DIV and one MOD.*

5 • FizzBuzz

the classic — every programmer has written this

Print the numbers 1 to 100, except:

- if the number divides by **3**, print Fizz instead
- if it divides by **5**, print Buzz instead
- if it divides by **both**, print FizzBuzz

The first fifteen lines should read:

```
1 2 Fizz 4 Buzz Fizz 7 8 Fizz Buzz 11 Fizz 13 14 FizzBuzz
```

The trap, and it catches nearly everybody. Most people write the test for 3 first, then the test for 5, then wonder why 15 prints Fizz instead of FizzBuzz.

Think about the *order* of your tests before you write them.

Stretch

6 • Leap years

A year is a leap year if it divides by **4** — *except* that century years are not, *unless* they divide by **400**.

So **2024** is a leap year. **1900** is not. **2000** is.

Write a program that asks for a year and says whether it is a leap year. **Three MODs.**

Test it with: 2024 · 2023 · 1900 · 2000.

7 • Squares and cubes

adapted from Paul Long

Using **:

(a) Ask for the length of a side and print the **area of the square**.

(b) Ask for the length of a side and print the **volume of the cube**.

(c) Ask for a radius and print the **volume of a sphere**. *You will need $V = \frac{4}{3}\pi r^3$ — and a value for π .*

8 • The same fact from both sides

Paper 1 meets Paper 2

You learned **binary shifts** for Paper 1 — shift left to multiply by 2, shift right to divide by 2. You have just learned DIV and MOD for Paper 2.

They are the same thing. Python writes a shift as >> and <<.

Operator	Shift	Same?
22 // 2	22 >> 1	
22 * 2	22 << 1	
4827 // 8	4827 >> 3	

Check all three at the machine. Then think about what MOD is doing:

4827 in binary is 1001011011011. Its last bit is 1, and $4827 \bmod 2$ is 1.

$n \bmod 2$ **is the last bit of n** . So “is this number even?” really means “is the last bit a 0?” — and $n \bmod 16$ is the last *four* bits, which is exactly one hexadecimal digit.

So why does the processor care?

A multiply or a divide is genuine work — the hardware has to compute it. **A shift is almost free:** nothing is calculated, the bits are simply wired one column across. It is the fastest thing a processor can do.

So when you write $n // 2$, a good compiler does not divide. **It shifts.** You wrote the Paper 2 operator; the machine ran the Paper 1 one.

This is not on the specification. It is here because the two halves of your course are describing the same machine, and it is worth seeing that once.

Everything on this page uses `DIV`, `MOD` or `^` and nothing else you have not already met. If you are stuck, the question is almost always “how many whole ones fit, and what is left over?”